CONSTRUCTION DOCUMENTS

Project:
WELD COUNTY JAIL MAU REPLACEMENT
2110 O Street
Greeley, Colorado 80631

Owner:
WELD COUNTY
1105 H Street
Greeley, Colorado 80631

Architects:
CAIRN DESIGN, LLC
1805 Sheely Drive
Fort Collins, Colorado 80526

Engineers:
CAIRN DESIGN, LLC
1805 Sheely Drive
Fort Collins, Colorado 80526

Upon request the county will make reasonable accommodations for alternative access to the listed documents. Please submit your request via Weld Request for Accommodation Form , email ADACoordinator@weld.gov, or by calling Human Resources at (970) 400-4234.

BUILDING CODE SUMMARY

GENERAL
The project consists of a roof top unit replacement.

The building is institutional occupancy and is sprinklered.

APPLICABLE CODES FOR REFERENCE. CONTRACTOR MUST VERIFY THE LATEST EDITION OF EACH CODE ADOPTED BY LOCAL GOVERNING AUTHORITIES.
- 2018 International Existing Building Code
- 2018 International Building Code
- 2018 International Mechanical Code
- 2018 International Plumbing Code
- 2018 International Fuel Gas Code
- 2018 International Energy Conservation Code
- 2023 National Electrical Code
- International Fire Code as adopted by Weld County
- Local Ordinances

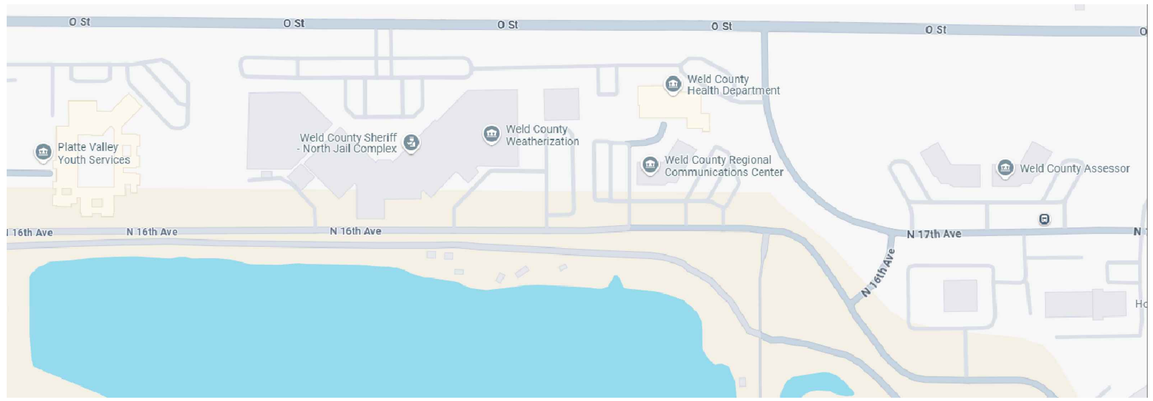
CHAPTER 3 - USE and OCCUPANCY CLASSIFICATION
USE AND OCCUPANCY CLASSIFICATION 303
Occupancy is institutional (I-3)

DRAWING INDEX

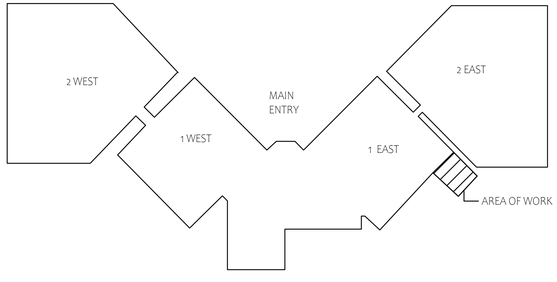
GENERAL / ARCHITECTURAL
A-01 COVER SHEET / BUILDING CODE SUMMARY
A-10 ARCHITECTURAL PARTIAL FLOOR PLANS AND DETAILS

MECHANICAL
M-0.1 MECHANICAL LEGEND AND NOTES
M-0.2 HVAC AND PLUMBING NOTES
M1.1 MECHANICAL AND PLUMBING PLAN

ELECTRICAL
E-01 SYMBOLS AND NOTES
E-10 ELECTRICAL PLAN



01 LOCATION PLAN
NO SCALE



02 KEYPLAN
NO SCALE

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1805 Sheely Drive
Fort Collins, CO 80526

voice: (970)-286-7968

cairndesignllc.com

STATE OF COLORADO
KATHERINE LIGON
404152
LICENSED ARCHITECT

PROJECT:

WELD COUNTY
JAIL NORTH
JAIL 1 EAST
MAKEUP AIR UNIT

SHEET TITLE:

COVER SHEET
BUILDING CODE
SUMMARY
-

ISSUE DATE:

BID/PERMIT
07/03/2026

REVISIONS:

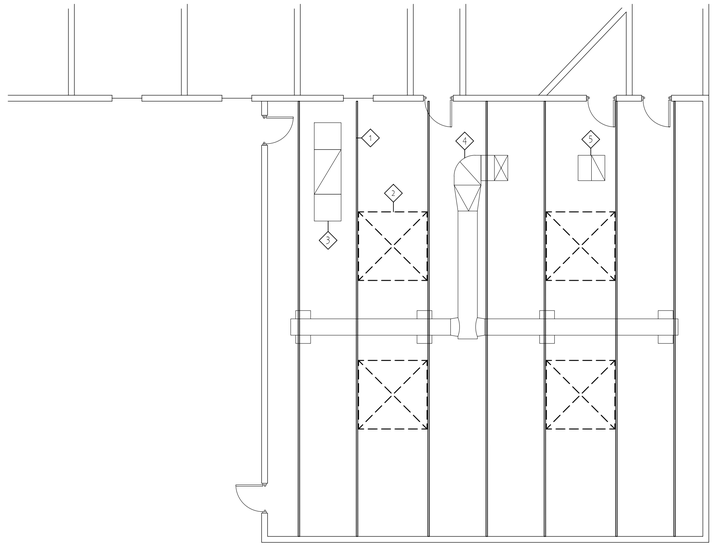
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PROJECT NO.:
2607

SHEET NO.

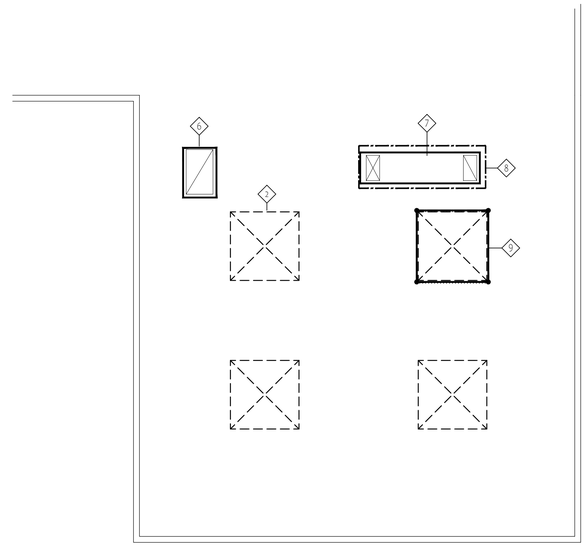
A-01

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01 PARTIAL FIRST FLOOR PLAN

1/8" = 1'-0"



02 PARTIAL ROOF PLAN

1/8" = 1'-0"

GENERAL NOTES:

1. THE CONTRACTOR SHALL INSPECT THE EXISTING CONDITIONS AND NOTE ANY ITEMS THAT NEED TO BE ADDRESSED, PRIOR TO THE START OF CONSTRUCTION.
2. ANY UNUSED PENETRATIONS SHALL BE CAPPED OFF AND MADE WATERTIGHT.

KEYED PLAN NOTES: ◇

1. EXISTING JOIST TO REMAIN. REFER TO STRUCTURAL LETTER ON THIS SHEET.
2. EXISTING SKYLIGHT TO REMAIN.
3. EXISTING RELIEF HOOD AND DUCTWORK TO BE REMOVED. REFER TO MECHANICAL.
4. EXISTING SUPPLY DUCT AND DIFFUSERS TO REMAIN.
5. EXISTING RETURN DUCT TO REMAIN.
6. PROVIDE INSULATED ROOF CAP OVER EXISTING CURB. MATCH ROOF INSULATION. INSTALL METAL DECK PATCH OVER EXISTING HOLE. PAINT TO MATCH EXISTING EXPOSED DECKING.
7. EXISTING RTU CURB TO REMAIN.
8. NEW CURB ADAPTOR. CONTRACTOR SHALL PROVIDE AN ADAPTOR PER FIELD MEASUREMENTS AND UNIT BEING PROVIDED.
9. PROVIDE SELF-SUPPORTING, FREESTANDING GUARD RAIL AROUND SKYLIGHT, GALVANIZED STEEL WITH BASEPLATES AND RUBBER PADS, OSHA COMPLIANT, UNISTRUT OR EQUAL.

Structural
Geotechnical

SINCE 1986 RMG 40 CELEBRATING YEARS

Materials Testing
Forensic

Job No. 201061

June 23, 2026

Cairn Design
1805 Sheely Drive
Fort Collins, CO 80526

Attn: Ken Caudle

Re: Roof Framing Review
Weld County Jail Renovations
2110 O St
Greeley, Colorado

Dear Ken:

As requested, personnel of RMG Engineers (RMG) completed a limited review of the framing at the address referenced above. The purpose of the review was to determine the structural adequacy of the existing roof framing to support new mechanical equipment. Reference the structural plans for the Weld County North Jail (Phase 2) by Kirkham Michael Consulting Engineers dated 6-12-2002, the existing mechanical plan by RMH Group dated 5-31-2002, and mechanical renovation plans by Evolution Consulting Group dated 5-8-2026. The anticipated loadings consist of a 30 pounds per square foot (psf) ground snow load, a 30 psf roof dead load, and a lateral load consisting of a 3 second gust at 112 miles-per-hour, exposure C, in accordance with IBC 2024, ASCE7-22, and the Weld County Local Amendments.

Personnel of Cairn Design have requested that RMG complete a review of the existing framing to determine if the existing roof framing is structurally sufficient to support new mechanical equipment. RTU-25 is to be removed and replaced with a new mechanical unit. In the existing mechanical plans noted above, RTU-25 is located on the roof of the southeast wing of the building. The existing unit and curb have a combined weight of approximately 1,580 lbs. According to the mechanical renovation plans noted above, the new unit will be installed in the same location as RTU-25 with a curb adapter. The combined weight of the new mechanical unit and the curb adapter is approximately 3,000 lbs. The existing structural plans noted above show the roof at RTU-25 to be framed with open web steel joists and 1.5B 18ga steel decking. The joists span in the north-south direction with a span length of approximately 49'-2". The typical center-to-center joist spacing is 6'-6", with one atypical spacing of 8'-0".

Personnel of Cairn Design completed a site visit to the address referenced above and provided field measurements of the steel joists to RMG via email on 5-27-2026. The field measurements were provided to The Steel Joist Institute who stated that the existing joists resemble a 36LH08 joist, assuming it was manufactured by a SJI member company.

It is our opinion that the existing roof framing is structurally sufficient to support the anticipated loads and new mechanical equipment noted above. The moment and shear reactions are increased by less than 5%

Southern Office:
Colorado Springs, CO 80919
719.548.0600

Central Office:
Englewood, CO 80112
303.688.9475
rmg-engineers.com

Northern Office:
Windsor, CO 80550
970.330.1071

when comparing the analysis results between the joists loaded with the existing RTU-25 and joists loaded with the proposed new mechanical unit and curb adapter.

Note, no framing other than that described above has been reviewed.

This report does not express nor does it imply any warranty of the future performance of the existing framing. No design calculations were performed on the existing framing by RMG unless noted. Not included is a review of architectural, mechanical, electrical, plumbing, or cosmetic conditions, nor a comprehensive review of compliance with applicable building codes.

I hope this provides the information you requested. Should you have any questions, please do not hesitate to call.

Cordially,

RMG Engineers

COLORADO LICENSED PROFESSIONAL ENGINEER 6/23/26 53501

Michael McGuire, P.E.
Structural Department Director

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1805 Sheely Drive
Fort Collins, CO 80526

voice: (970)-286-7968

cairndesignllc.com

STATE OF COLORADO KATHERINE LIGON 404152 LICENSED ARCHITECT

PROJECT:

WELD COUNTY
JAIL NORTH
JAIL 1 EAST
MAKEUP AIR UNIT

SHEET TITLE:

PARTIAL FLOOR PLAN
AND DETAILS

-

ISSUE DATE:

BID/PERMIT
07/03/2026

REVISIONS:

-
-
-
-

PROJECT NO.:
2607

SHEET NO.

A-10

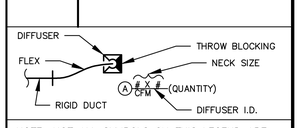
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MECHANICAL LEGEND

SYMBOL	ABBREVIATION	DESCRIPTION
———	(E)	EXISTING (LIGHT LINE)
×××××	DEMO	TO BE DEMOLISHED
———	(N)	NEW (HEAVY LINE)
—CHR—	CHR	DUAL TEMPERATURE RETURN
—CHS—	CHS	DUAL TEMPERATURE SUPPLY
—CR—	CR	CHILLED WATER RETURN
—CS—	CS	CHILLED WATER SUPPLY
—CWR—	CWR	CONDENSER WATER RETURN
—CWS—	CWS	CONDENSER WATER SUPPLY
—HWR—	HWR	HEATING WATER RETURN
—HWS—	HWS	HEATING WATER SUPPLY
—HPC—	HPC	HIGH PRESSURE CONDENSATE
—HPS—	HPS	HIGH PRESSURE STEAM
—LPC—	LPC	LOW PRESSURE CONDENSATE
—LPS—	LPS	LOW PRESSURE STEAM
—MPC—	MPC	MEDIUM PRESSURE CONDENSATE
—MPS—	MPS	MEDIUM PRESSURE STEAM
—PC—	PC	PUMPED CONDENSATE
—RD—	RD	REFRIGERANT DISCHARGE
—RL—	RL	REFRIGERANT LIQUID
—RS—	RS	REFRIGERANT SUCTION
— — —	CW	COLD WATER
— — — —	HW	HOT WATER
— — — —	HWC	HOT WATER CIRCULATION
—A—	A	AIR PIPING (COMPRESSED)
—D—	D	DRAIN
—DIC—	DIC	DE-IONIZED WATER CIRCULATION
—F—	F	FIRE
—FOR—	FOR	FUEL OIL RETURN
—FOS—	FOS	FUEL OIL SUPPLY
—FOV—	FOV	FUEL OIL VENT
—G—	G	GAS
—GW—	GW	GREASE WASTE
—MA—	MA	MEDICAL AIR
—N—	N	NITROGEN
—ORDL—	ORDL	OVERFLOW ROOF DRAIN LEADER
—O—	OXY	OXYGEN
—RDL—	RDL	ROOF DRAIN LEADER
—SAN—	SAN	SANITARY WASTE
—SOD—	SOD	SAND AND OIL DRAIN
—ST—	ST	STORM
—VAC—	VAC	VACUUM
—V—	V	VENT
—W—	W	WASTE

SYMBOL	DESCRIPTION
	PIPE UP AND DOWN ELBOW
	PIPE DOWN TEE
	PIPE CONTINUATION
	PIPE REDUCER
	PIPE UNION AND CAP
	BALL VALVE, GATE VALVE
	MOTORIZED VALVE – 2–WAY
	MOTORIZED VALVE – 3–WAY
	PRESSURE REDUCING VALVE
	CHECK VALVE
	GLOBE VALVE
	BALANCE VALVE
	STRAINER
	HOSE–END DRAIN VALVE
	STRAINER WITH BLOW–OFF
	VENTURI
	GAS COCK
	PUMP
	DUAL CHECK BACKFLOW PREVENTOR
	REDUCED PRESSURE BACKFLOW PREVENTOR
	PRESSURE–TEMPERATURE TAP
	PRESSURE–TEMPERATURE RELIEF VALVE
	AIR VENT
	PIPE GUIDE OR SLEEVE
	PIPE EXPANSION JOINT
	PIPE ANCHOR
	VACUUM BREAKER
	THERMOMETER
	PRESSURE GAUGE
FS	FLOW SENSOR
	FLOOR DRAIN
	FLOOR SINK–FULL–3/4–HALF GRATE
	DRAIN ABOVE
	ROOF DRAIN
	ROOF DRAIN – OVERFLOW
	DOWNSPOUT NOZZLE
	CLEANOUT – VERTICAL
	CLEANOUT – HORIZONTAL
W H	WALL HYDRANT, HOSE BIBB

SYMBOL	DESCRIPTION
	SMOKE DETECTOR
20/16	DUCT SIZE INDICATING SHEET METAL DIMENSIONS. FIRST NUMBER WIDTH AND SECOND IS DEPTH.
	DUCT ELBOW WITH TURNING VANE
	DUCT TEE WITH TURNING VANES
	MANUAL DAMPER WITH LOCKING QUADRANT.
M	MOTORIZED DAMPER
	FLEXIBLE DUCT CONNECTOR
	SPIN–IN FITTING WITH DAMPER
	45° DUCT TAKE–OFF
U.C. SIZE	DOOR UNDERCUT
	FIRE DAMPER
	FIRE AND SMOKE DAMPER
	SMOKE DAMPER
	EXISTING FIRE DAMPER
	SUPPLY DIFFUSER
	EXHAUST, RETURN, OR TRANSFER GRILLE
CO	CARBON MONOXIDE SENSOR
T XXX	THERMOSTAT OR TEMPERATURE SENSOR FOR DEVICE "XXX"
	CONNECTION NEW TO EXISTING



NOTE: NOT ALL SYMBOLS ON THIS LEGEND ARE NECESSARILY USED ON THIS PROJECT.

SPECIFICATION GENERAL NOTES

GENERAL NOTES:

A. REVIEW THE CONTRACT CONDITIONS AND GENERAL REQUIREMENTS FOR INFORMATION THAT APPLIES.

B. THE WORD "PROVIDE" IS USED TO MEAN "FURNISH AND INSTALL."

C. PROVIDE ALL ITEMS FOR A COMPLETE AND SUCCESSFUL OPERATION OF ALL SYSTEMS SHOWN ON THESE DRAWINGS.

D. THE DRAWINGS ARE DIAGRAMMATIC AND INDICATE GENERAL ARRANGEMENT OF THE WORK. LOCATIONS ARE APPROXIMATE UNLESS DIMENSIONED. MINOR MODIFICATIONS IN LOCATION TO MEET SITE REQUIREMENTS ARE ACCEPTABLE. REFER TO LATEST ARCHITECTURAL DRAWINGS FOR EXACT WALL LOCATIONS AND DIMENSIONS.

E. LAY OUT ALL WORK IN ADVANCE. DO NOT DEFACE THE WORK OF OTHER TRADES OR THE EXISTING BUILDING.

F. REPAIR ALL WALL, FLOOR, AND CEILING PENETRATIONS TO MAINTAIN THE EXISTING FIRE RATING.

G. THE CONTRACTOR SHALL VISIT THE SITE PRIOR TO BIDDING THE PROJECT AND TAKE INTO CONSIDERATION CONDITIONS THAT MAY AFFECT THE WORK. NO ADDITIONAL COMPENSATION WILL BE APPROVED FOR FAILURE TO VISIT THE SITE PRIOR TO PRICING THE WORK.

H. MAINTAIN A CONTRACT SET OF THESE DRAWINGS AT THE SITE, WITH ALL CHANGES OR DEVIATIONS FROM THE ORIGINAL DRAWINGS NEATLY MARKED ON THEM IN RED COLOR. THIS SHALL BE A SEPARATE SET OF DRAWINGS NOT USED FOR CONSTRUCTION PURPOSES, WHICH SHALL BE KEPT UP TO DATE AS THE JOB PROGRESSES AND SHALL BE MADE AVAILABLE FOR INSPECTION BY THE ENGINEER AT ALL TIMES. UPON COMPLETION OF THE CONTRACT, THIS SET OF "AS–BUILTS" SHALL BE DELIVERED TO THE ENGINEER.

I. MATERIAL SHALL BE AS SPECIFIED. SUBSTITUTIONS WILL BE CONSIDERED IF SUBMITTED AT LEAST ONE (1) WEEK PRIOR TO THE CONTRACT BID DATE. SUBSTITUTIONS SHOULD BE SUBMITTED SEPARATELY FOR EACH PRODUCT WITH SUPPORTING DATA, DRAWINGS AND SAMPLES AS APPROPRIATE, INCLUDING: 1.) COMPARISON OF THE QUALITIES OF THE PROPOSED SUBSTITUTION WITH THAT SPECIFIED. 2.) CHANGES REQUIRED IN OTHER ELEMENTS OF THE WORK BECAUSE OF THE SUBSTITUTION. 3.) COST DATA COMPARING THE PROPOSED SUBSTITUTION WITH THE PRODUCT SPECIFIED. THE ENGINEER WILL DETERMINE THE ACCEPTABILITY OF THE PROPOSED SUBSTITUTION.

J. SUBMIT THREE (3) COPIES OF MANUFACTURER'S SHOP DRAWINGS FOR EQUIPMENT AND DEVICES. PRIOR TO SUBMITTING THE SHOP DRAWINGS FOR REVIEW, THE CONTRACTOR SHALL REVIEW AND CERTIFY SAME AS TO COMPLIANCE WITH THE PLANS AND SPECIFICATIONS AND FOR DIMENSIONAL SUITABILITY FOR THE APPLICATIONS.

K. THE CONTRACTOR SHALL PROVIDE ALL EQUIPMENT AND PERSONNEL REQUIRED FOR TESTING OF INSTALLED EQUIPMENT.

I. THE CONTRACTOR SHALL DEMONSTRATE THE PROPER OPERATION OF ALL SYSTEMS TO THE OWNER, AT A TIME AS AGREED TO BY THE OWNER AND DIRECTED BY THE OWNER.

J. THE CONTRACTOR SHALL INCLUDE THE COST FOR A TEST, ADJUST, BALANCE CONTRACTOR TO VERIFY THE PERFORMANCE OF THE SYSTEMS SHOWN ON THIS DRAWING AND SET ALL SYSTEM AIR AND WATER FLOWS. TAB CONTRACTOR SHALL BE CERTIFIED BY THE ASSOCIATED AIR BALANCE COUNCIL (AABC) OR THE NATIONAL ENVIRONMENTAL BALANCING BUREAU (NEBB). SUBMIT TO ENGINEER FOR REVIEW.

M. THE CONTRACTOR SHALL INCLUDE THE COST FOR COMMISSIONING TO VERIFY THE PERFORMANCE OF THE SYSTEM CONTROLS SHOWN ON THIS DRAWING. THIS INCLUDES PREPARING A COMMISSIONING PLAN, VERIFYING THE OPERATION OF ALL SYSTEMS PER CODE REQUIREMENTS, AND PROVIDING A FINAL COMMISSIONING REPORT. SUBMIT TO ENGINEER FOR REVIEW.

CODES, REGULATIONS, AND STANDARDS:

1. ALL WORK SHALL BE IN STRICT ACCORD WITH LOCAL GOVERNING LAWS, ORDINANCES, AND REGULATIONS; ALL WORK MUST BE IN FULL ACCORDANCE WITH ALL CODES, ORDINANCES, AND CODE RULINGS. CONTRACTOR SHALL PROVIDE, WITHOUT EXTRA CHARGE, THE LABOR AND MATERIALS REQUIRED FOR FULL CODE COMPLIANCE.
2. ALL MATERIALS SHALL BE NEW AND SHALL COMPLY WITH THE SPECIFICATIONS ON DRAWINGS.
3. CONTRACTOR SHALL OBTAIN AND PAY FOR ALL LOCAL FEES, PERMITS, AND SERVICES OF INSPECTION AUTHORITIES REQUIRED BY THE WORK OF THE PROJECT. THE CONTRACTOR SHALL ARRANGE FOR ALL INSPECTIONS WHEN THEY BECOME DUE AND SHALL NOT COVER ANY WORK UNTIL APPROVED BY THE INSPECTION AUTHORITY.

GUARANTEE:

A. THE CONTRACTOR SHALL GUARANTEE ALL WORKMANSHIP, MATERIALS, AND EQUIPMENT PROVIDED FOR THE PROJECT AGAINST DEFECTS AND/OR FAULTY WORKMANSHIP FOR A PERIOD OF THREE HUNDRED SIXTY FIVE (365) CALENDAR DAYS FROM THE DATE OF FINAL ACCEPTANCE BY THE OWNER.

B. THE CONTRACTOR SHALL REPAIR AND/OR REPLACE DEFECTIVE OR FAULTY WORKMANSHIP, MATERIALS, OR EQUIPMENT, AND SHALL BE RESPONSIBLE FOR THE REPAIR AND/OR REPLACEMENT OF OTHER PROPERTY OR WORK DAMAGED AS A RESULT, WITHOUT CHARGE TO THE OWNER AND AS QUICKLY AS POSSIBLE, DURING THE GUARANTEE PERIOD.

PLUMBING GENERAL NOTES

1. THE NEW PLUMBING SYSTEMS SHALL BE BY THE PLUMBING CONTRACTOR. THIS INCLUDES, BUT IS NOT LIMITED TO:
   1.1. THE GAS DISTRIBUTION SYSTEMS WITH: GAS METER SERVICE ENTRANCE, PRESSURE REGULATION, PIPING, SUPPORTS, VALVES, EQUIPMENT CONNECTIONS, CONTROL SYSTEMS, ETC.
2. ALL PLUMBING SHALL BE IN ACCORDANCE WITH THE LOCAL PLUMBING CODES AND/OR ORDINANCES, INCLUDING, BUT NOT LIMITED TO PIPE SIZE.
3. GAS PIPING SHALL BE CODE APPROVED MATERIALS FOR PIPING AND FITTINGS.
4. PROVIDE LINE SIZE SHUT–OFF VALVE, REGULATOR, MALLEABLE UNION, AND DIRT LEG ON ALL GAS EQUIPMENT.
5. ALL MATERIAL AND EQUIPMENT PROVIDED AND INSTALLED UNDER THIS SECTION SHALL BE NEW AND IN CLEAN AND BRIGHT CONDITION. THE CONTRACTOR SHALL TAKE ANY MEASURE NECESSARY TO ENSURE AND MAINTAIN THE QUALITY OF THE INSTALLATION. PROVIDE FREEZE PROTECTION FOR PIPING WITHIN UNHEATED SPACES.
6. ALL PIPING IS TO BE TESTED IN ACCORDANCE WITH ACCEPTED CODES OF PRACTICE.
7. ALL GAS REGULATORS SHALL BE VENTED TO EXTERIOR OF THE BUILDING.
12. PROVIDE ALL CURBS, SUPPORTS, AND ANCHORS FOR MECHANICAL WORK. NO CHAIN, TAPE OR WIRE MAY BE USED FOR HANGING OR SUPPORTING.
13. ALL EXTERIOR GAS PIPING SHALL BE CLEANED AND PAINTED WITH EXTERIOR GRADE PAINT TO PROTECT AGAINST CORROSION. COORDINATE DESIGN COLOR WITH ARCHITECT.
14. RECEIVED, UNCRATED, ASSEMBLE, INSURE, AND INSTALL IN CONFORMANCE TO MANUFACTURER'S RECOMMENDATIONS, ALL EQUIPMENT FURNISHED BY THIS CONTRACT AND FURNISHED BY THE OWNER.
15. THIS CONTRACTOR SHALL COORDINATE ALL PIPING PRIOR TO INSTALLATION.
9. WEATHERPROOF ALL ROOF PENETRATIONS PER CODES AND ALL ROOFING MANUFACTURER RECOMMENDATIONS.
10. PROVIDE SLEEVES AND COLLARS FOR ALL PIPES THROUGH WALL, FLOORS, AND CEILINGS. SEAL ALL EXTERNAL PENETRATIONS OF FIRE RATED WALL, FLOORS, AND CEILINGS.
11. A MINIMUM CLEARANCE OF 30 INCHES SHALL BE PROVIDED AROUND ANY EQUIPMENT (I.E., FANS, PUMPS, BOILERS, AIR CONDITIONERS, ETC.) FOR SERVICE AND MAINTENANCE.

MECHANICAL DRAWING INDEX

DRAWING NUMBER	DRAWING TITLE	ISSUE DATE – DESCRIPTION: 07–03–2026 – BID/PERMIT SET
M0.1	MECHANICAL LEGEND AND NOTES	●
M0.2	HVAC AND PLUMBING NOTES AND CONTROLS	●
M1.0	MECHANICAL AND PLUMBING PLAN	●

KEY:
● FULL SHEET ISSUE.
Ⓟ PARTIAL SHEET ISSUE (SKETCH).

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BUILDING CODE DATA

DISCIPLINE	CODE	EDITION
GENERAL	INTERNATIONAL BUILDING CODE	2018
HVAC / MECHANICAL	INTERNATIONAL MECHANICAL CODE	2018
PLUMBING	INTERNATIONAL PLUMBING CODE	2018
ELECTRICAL	NATIONAL ELECTRIC CODE	2023
FIRE PROTECTION	INTERNATIONAL FIRE CODE	2018
	NFPA STANDARDS	CURRENT
FUEL	INTERNATIONAL FUEL GAS CODE	2018
ENERGY	INTERNATIONAL ENERGY CONSERVATION CODE	2018
ACCESSIBILITY	INTERNATIONAL BUILDING CODE / ANSI	2018

NOTES:
A. ALL WORK PERFORMED SHALL COMPLY WITH THE REQUIREMENTS OF THE LISTED CODES, LOCAL CODE AMENDMENTS, AND REFERENCED STANDARDS AS ENFORCED BY THE AUTHORITY HAVING JURISDICTION (AHJ).
B. ALL WORK SUBJECT TO INSPECTION BY THE AHJ AT THE PROJECT SITE FOR COMPLIANCE. ENSURE ALL LOCAL CODE AMENDMENTS ARE ALSO FOLLOWED.

HVAC GENERAL NOTES

1. THE NEW MECHANICAL SYSTEMS SHALL BE BY THE MECHANICAL CONTRACTOR. THIS INCLUDES, BUT IS NOT LIMITED TO:
   1.1. THE AIR DISTRIBUTION SYSTEMS WITH: AIR HANDLING EQUIPMENT, DUCTWORK, CONTROL SYSTEMS, ETC.
2. AIR BALANCE SHALL BE BY THE CONTRACTOR IN ACCORDANCE WITH THE CONTRACT DRAWINGS. PROVIDE NEBB CERTIFIED REPORT TO THE OWNER (3 SETS).
3. IDENTIFY ALL HVAC EQUIPMENT AS TO THE AREA SERVED BY THE EQUIPMENT. IDENTIFICATION SHALL BE ENGRAVED PLASTIC TAGS PERMANENTLY AFFIXED TO EACH PIECE OF EQUIPMENT.
4. PROVIDE ALL CURBS, SUPPORTS, AND ANCHORS FOR MECHANICAL WORK. NO CHAIN, TAPE OR WIRE MAY BE USED FOR HANGING OR SUPPORTING.
5. RECEIVED, UNCRATED, ASSEMBLE, INSURE, AND INSTALL IN CONFORMANCE TO MANUFACTURER'S RECOMMENDATIONS, ALL EQUIPMENT FURNISHED BY THIS CONTRACT AND FURNISHED BY THE OWNER.
6. DUCT DIMENSIONS ARE CLEAR INSIDE DIMENSIONS. OVERALL OUTSIDE DUCT DIMENSIONS SHALL BE ADJUSTED TO ALLOW FOR ANY LINEAR THICKNESS.
7. ALL SHEET METAL TO BE MADE AND INSTALLED TO SMACNA STANDARDS WITH 45 DEGREE MAXIMUM REDUCING, 30 DEGREE MAXIMUM EXPANDING TRANSITIONS. SEAL ALL DUCT AIRTIGHT WITH TWO COATS OF DUCT SEALANT. DUCT SEALANT TO BE CLEAR PAINTABLE SILICON CAULK ON ALL EXPOSED DUCTWORK.
8. SUPPORT ALL DUCTWORK FROM STRUCTURE. EQUIPMENT SHALL NOT BE USED TO SUPPORT DUCTWORK. PROVIDE DUCT SUPPORTS PER IMC REQUIREMENTS AND SMACNA GUIDELINES / RECOMMENDATIONS.
9. CAULK ALL DUCT JOINTS AIR AND WATER TIGHT WITH PERMANENT COMMERCIAL CAULK PER MANUFACTURER'S RECOMMENDATIONS.
10. THIS CONTRACTOR SHALL COORDINATE ALL DUCTWORK PRIOR TO INSTALLATION.
11. WEATHERPROOF ALL MECHANICAL ROOF PENETRATIONS PER CODES AND ALL ROOFING MANUFACTURER RECOMMENDATIONS.
12. PROVIDE SLEEVES AND COLLARS FOR ALL DUCTWORK AND PIPES THROUGH WALL, FLOORS, AND CEILINGS. SEAL ALL EXTERNAL PENETRATIONS OF FIRE RATED WALL, FLOORS, AND CEILINGS.
13. CONFIRM VOLTAGE, PHASE, AND AMPACITY WITH ELECTRICAL CONTRACTOR PRIOR TO ORDERING EQUIPMENT. ALL CONTROL AND INTERLOCK WIRING FOR MECHANICAL EQUIPMENT BY MECHANICAL CONTRACTOR. THREE–PHASE MOTORS TO HAVE MAGNETIC STARTERS WITH PROTECTION ON ALL THREE LEADS. CONTROL AND HEATING/COOLING EQUIPMENT TO AUTOMATICALLY RESTART AFTER POWER FAILURE. ALL WITH IN CONDUIT AS REQUIRED BY CODE.
14. A MINIMUM CLEARANCE OF 30 INCHES SHALL BE PROVIDED AROUND ANY EQUIPMENT (I.E., FANS, PUMPS, BOILERS, ETC.) FOR SERVICE AND MAINTENANCE UNLESS MANUFACTURER REQUIREMENTS DICTATE OTHERWISE.

CAIRN DESIGN ENGINEERS
ARCHITECTS
leading high performance building solutions

leading high performance building solutions

MWBE - DBE - SBE

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Fort Collins, CO 80526

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COLORADO REGISTERED
JENNIFER C. DAVIDSON
NO. 33293
07/03/26
PROFESSIONAL ENGINEER

PROJECT:
WELD COUNTY
JAIL NORTH
JAIL 1 EAST
MAKEUP AIR UNIT

SHEET TITLE:
MECHANICAL LEGEND AND NOTES

ISSUE DATE:
BID/PERMIT
07/03/2026

REVISIONS:

PROJECT NO.:
2607

SHEET NO.
M0.1

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SEQUENCES OF OPERATION

1. ROOFTOP UNIT:

THE NEW ROOFTOP UNIT SYSTEM IS TO BE PACKAGED CONTROLS WITH INTERFACE TO THE EXISTING BAS SYSTEMS BY DELTA CONTROLS VERSION 3.4. TREND ALL POINTS AND ALARMS. UNIT IS NOT UNDER EMERGENCY POWER CONTROL. RETAIN ANY SMOKE MANAGEMENT AND ELECTRICAL DEMAND LOAD SHEDDING SEQUENCES WHICH MAY EXIST.

1.1 SINGLE ZONE UNIT

RUN CONDITIONS – SCHEDULED:
THE UNIT SHALL RUN ACCORDING TO A USER DEFINABLE TIME SCHEDULE IN THE FOLLOWING MODES:
OCCUPIED MODE: THE UNIT SHALL MAINTAIN
A 75°F (ADJ.) COOLING SETPOINT
A 70°F (ADJ.) HEATING SETPOINT.
UNOCCUPIED MODE (NIGHT SETBACK): THE UNIT SHALL MAINTAIN
A 85°F (ADJ.) COOLING SETPOINT.
A 55°F (ADJ.) HEATING SETPOINT.

ALARMS SHALL BE PROVIDED AS FOLLOWS:
HIGH ZONE TEMP: IF THE ZONE TEMPERATURE IS GREATER THAN THE COOLING SETPOINT BY A USER DEFINABLE AMOUNT (ADJ.).
LOW ZONE TEMP: IF THE ZONE TEMPERATURE IS LESS THAN THE HEATING SETPOINT BY A USER DEFINABLE AMOUNT (ADJ.).

ZONE SETPOINT ADJUST:
THE OCCUPANT SHALL BE ABLE TO ADJUST THE ZONE TEMPERATURE HEATING AND COOLING SETPOINTS AT THE ZONE SENSOR.

ZONE OPTIMAL START:
THE UNIT SHALL USE AN OPTIMAL START ALGORITHM FOR MORNING START–UP. THIS ALGORITHM SHALL MINIMIZE THE UNOCCUPIED WARM–UP OR COOL–DOWN PERIOD WHILE STILL ACHIEVING COMFORT CONDITIONS BY THE START OF SCHEDULED OCCUPIED PERIOD.

ZONE UNOCCUPIED OVERRIDE:
A TIMED LOCAL OVERRIDE CONTROL SHALL ALLOW AN OCCUPANT TO OVERRIDE THE SCHEDULE AND PLACE THE UNIT INTO AN OCCUPIED MODE FOR AN ADJUSTABLE PERIOD OF TIME. AT THE EXPIRATION OF THIS TIME, CONTROL OF THE UNIT SHALL AUTOMATICALLY RETURN TO THE SCHEDULE.

FREEZE PROTECTION:
THE UNIT SHALL SHUT DOWN AND GENERATE AN ALARM UPON RECEIVING A FREEZESTAT STATUS.

RETURN AIR SMOKE DETECTION:
THE UNIT SHALL SHUT DOWN AND GENERATE AN ALARM UPON RECEIVING A RETURN AIR SMOKE DETECTOR STATUS.

SUPPLY AIR SMOKE DETECTION:
THE UNIT SHALL SHUT DOWN AND GENERATE AN ALARM UPON RECEIVING A SUPPLY AIR SMOKE DETECTOR STATUS.

SUPPLY FAN:
THE SUPPLY FAN SHALL RUN ANYTIME THE UNIT IS COMMANDED TO RUN, UNLESS SHUTDOWN ON SAFETIES. TO PREVENT SHORT CYCLING, THE SUPPLY FAN SHALL HAVE A USER DEFINABLE (ADJ.) MINIMUM RUNTIME. ALARMS SHALL BE PROVIDED AS FOLLOWS:
SUPPLY FAN FAILURE: COMMANDED ON, BUT THE STATUS IS OFF.
SUPPLY FAN IN HAND: COMMANDED OFF, BUT THE STATUS IS ON.
SUPPLY FAN RUNTIME EXCEEDED: STATUS RUNTIME EXCEEDS A USER DEFINABLE LIMIT (ADJ.).

ZONE TEMPERATURE CONTROL:
THE CONTROLLER SHALL MEASURE THE ZONE TEMPERATURE AND SHALL MODULATE THE SUPPLY FAN VFD SPEED TO MAINTAIN ZONE TEMPERATURE SETPOINT. THE FAN SPEED SHALL INCREASE AS THE ZONE TEMPERATURE RISES ABOVE COOLING SETPOINT, OR AS THE ZONE TEMPERATURE DROPS BELOW HEATING SETPOINT. THE SUPPLY FAN VFD SPEED SHALL NOT DROP BELOW 30% (ADJ.).

COOLING STAGES:
THE CONTROLLER SHALL MEASURE THE ZONE TEMPERATURE AND STAGE THE COOLING TO MAINTAIN ITS COOLING SETPOINT. TO PREVENT SHORT CYCLING, THERE SHALL BE A USER DEFINABLE (ADJ.) DELAY BETWEEN STAGES,
AND EACH STAGE SHALL HAVE A USER DEFINABLE (ADJ.) MINIMUM RUNTIME.
THE COOLING SHALL BE ENABLED WHENEVER:
OUTSIDE AIR TEMPERATURE IS GREATER THAN 60°F (ADJ.).
AND THE ECONOMIZER (IF PRESENT) IS DISABLED OR FULLY OPEN.
AND THE ZONE TEMPERATURE IS ABOVE COOLING SETPOINT.
AND THE SUPPLY FAN STATUS IS ON.
AND THE HEATING IS NOT ACTIVE.

GAS HEATING STAGES:
THE CONTROLLER SHALL MEASURE THE ZONE TEMPERATURE AND STAGE THE HEATING TO MAINTAIN ITS HEATING SETPOINT. TO PREVENT SHORT CYCLING, THERE SHALL BE A USER DEFINABLE (ADJ.) DELAY BETWEEN STAGES, AND EACH STAGE SHALL HAVE A USER DEFINABLE (ADJ.) MINIMUM RUNTIME.
THE HEATING SHALL BE ENABLED WHENEVER:
OUTSIDE AIR TEMPERATURE IS LESS THAN 65°F (ADJ.).
AND THE ZONE TEMPERATURE IS BELOW HEATING SETPOINT.
AND THE SUPPLY FAN STATUS IS ON.
AND THE COOLING IS NOT ACTIVE.

ECONOMIZER:
THE CONTROLLER SHALL MEASURE THE ZONE TEMPERATURE AND MODULATE THE ECONOMIZER DAMPERS IN SEQUENCE TO MAINTAIN A SETPOINT 2°F LESS THAN THE ZONE COOLING SETPOINT. THE OUTSIDE AIR DAMPERS SHALL MAINTAIN A MINIMUM OPEN ADJUSTABLE POSITION TO MEET VENTILATION REQUIREMENTS SHOWN ON THESE DRAWINGS WHENEVER OCCUPIED.
THE ECONOMIZER SHALL BE ENABLED WHENEVER:
OUTSIDE AIR TEMPERATURE IS LESS THAN 75°F (ADJ.).
AND THE OUTSIDE AIR TEMPERATURE IS LESS THAN THE RETURN AIR TEMPERATURE.
AND THE SUPPLY FAN STATUS IS ON.
THE ECONOMIZER SHALL CLOSE WHENEVER:
MIXED AIR TEMPERATURE DROPS FROM 45°F TO 40°F (ADJ.).
OR ON LOSS OF SUPPLY FAN STATUS.
OR THE FREEZESTAT (IF PRESENT) IS ON.
THE OUTSIDE AIR DAMPERS SHALL CLOSE AND THE RETURN AIR DAMPER SHALL OPEN WHEN THE UNIT IS OFF. IF OPTIMAL START UP IS AVAILABLE, THE MIXED AIR DAMPER SHALL OPERATE AS DESCRIBED IN THE OCCUPIED MODE EXCEPT THAT THE OUTSIDE AIR DAMPER SHALL MODULATE TO FULLY CLOSED.

PREFILTER DIFFERENTIAL PRESSURE MONITOR:
THE CONTROLLER SHALL MONITOR THE DIFFERENTIAL PRESSURE ACROSS THE PREFILTER.

ALARMS SHALL BE PROVIDED AS FOLLOWS:
PREFILTER CHANGE REQUIRED: PREFILTER DIFFERENTIAL PRESSURE EXCEEDS A USER DEFINABLE LIMIT (ADJ.).

MIXED AIR TEMPERATURE:
THE CONTROLLER SHALL MONITOR THE MIXED AIR TEMPERATURE AND USE AS REQUIRED FOR ECONOMIZER CONTROL (IF PRESENT) OR PREHEATING CONTROL (IF PRESENT).

ALARMS SHALL BE PROVIDED AS FOLLOWS:
HIGH MIXED AIR TEMP: IF THE MIXED AIR TEMPERATURE IS GREATER THAN 90°F (ADJ.).
LOW MIXED AIR TEMP: IF THE MIXED AIR TEMPERATURE IS LESS THAN 45°F (ADJ.).

RETURN AIR TEMPERATURE:
THE CONTROLLER SHALL MONITOR THE RETURN AIR TEMPERATURE AND USE AS REQUIRED FOR ECONOMIZER CONTROL (IF PRESENT).

ALARMS SHALL BE PROVIDED AS FOLLOWS:
HIGH RETURN AIR TEMP: IF THE RETURN AIR TEMPERATURE IS GREATER THAN 90°F (ADJ.).
LOW RETURN AIR TEMP: IF THE RETURN AIR TEMPERATURE IS LESS THAN 45°F (ADJ.).

SUPPLY AIR TEMPERATURE:
THE CONTROLLER SHALL MONITOR THE SUPPLY AIR TEMPERATURE.

ALARMS SHALL BE PROVIDED AS FOLLOWS:
HIGH SUPPLY AIR TEMP: IF THE SUPPLY AIR TEMPERATURE IS GREATER THAN 120°F (ADJ.).
LOW SUPPLY AIR TEMP: IF THE SUPPLY AIR TEMPERATURE IS LESS THAN 45°F (ADJ.).

ROOF TOP UNIT SCHEDULE

TAG	MANUFACTURER/ MODEL (BASIS OF DESIGN)	SERVICE	OPER. ELEV. (FEET)	SUPPLY FAN DATA			MIN. OUTSIDE AIR FLOW (CFM)	RETURN FAN DATA			UNIT ELECTRICAL DATA			HEATING			COOLING			UNIT WEIGHT (LBS.)	NOTES
				AIR FLOW (CFM)	T.S.P. (IN. W.C.)	HORSE POWER		AIR FLOW (CFM)	E.S.P. (IN. W.C.)	HORSE POWER	VOLTAGE/ PHASES	MIN. CIRCUIT AMPS	MAX. OCP	FUEL OR FLUID	INPUT (MBH)	% EFF.	REF.	NOMINAL TONS	IEER/EER		
RTU–1	TRANE YSK180A4S0H	GYM	4,700	6000	1.2	3	890	NA	NA	NA	460/3	43	50	GAS	315	81	R454B	15.0	14.5/10.8	2377	A,B,C,D,E,F,G,H,I,J,K

NOTES:
A. WITH SUPPLY AND RETURN AIR SMOKE DETECTOR.
B. WITH ECONOMIZER (DRYBULB CONTROL).
C. WITH BAROMETRIC RELIEF.
D. WITH HAIL GUARDS.
E. WITH ADAPTOR CURB TO CONNECT NEW UNIT TO EXISTING CURB.
F. WITH THROUGH THE BASE ELECTRICAL CONNECTION.
G. WITH NON–FUSED DISCONNECT.
H. WITH POWERED CONVENIENCE OUTLET.
I. WITH 3 STAGE COOLING, 2 STAGE GAS HEAT.
J. WITH VAV SUPPLY FAN.
K. WITH PACKAGED UNIT CONTROLS SYMBIO 700 WITH BACNET BAS INTERFACE.

EQUALS BY: CARRIER, DIAKIN.

Ventilation Calculations

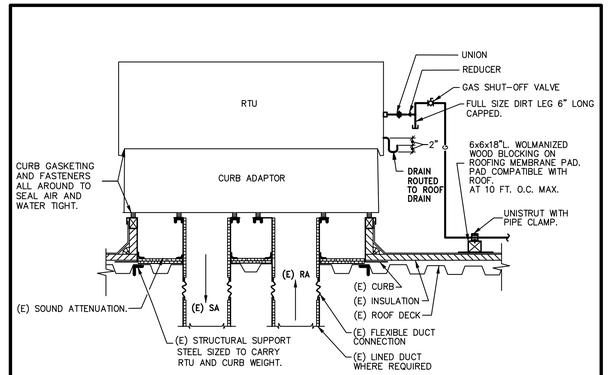
Room Name	Room Area sqft	Room Type	Number of People*	OA CFM/Person	OA CFM/sqft	Total Required OA CFM	Total Provided OA CFM
RTU-1							
Gym	2500	Gym, stadium, arena (play area)	22	20	0.18	890	890
Total						890	890

* per owner max occupancy

GAS LOAD SCHEDULE

TAG	DESCRIPTION	INPUT EACH (SEA LEVEL MBH)	QUANTITY	TOTAL INPUT (SEA LEVEL MBH)	INLET PRESSURE	NOTES
RTU–1	AIR HANDLING UNIT	315	1	315	7"	A
HVAC SUBTOTAL:				315 MBH		

NOTES:
A. EXISTING GAS LOAD REMOVED IS 330 MBH. EXISTING PIPE SIZE SUFFICIENT TO SERVE NEW UNIT. CONFIRM GAS PRESSURE.



RTU MOUNTING & GAS CONNECTION DETAIL
SCALE: NO SCALE

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COLORADO REGISTERED
CHRISTOPHER C. DAVIDSON
33293
07/03/26
PROFESSIONAL ENGINEER

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JAIL 1 EAST
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DEMOLITION WORK NOTES: (X)

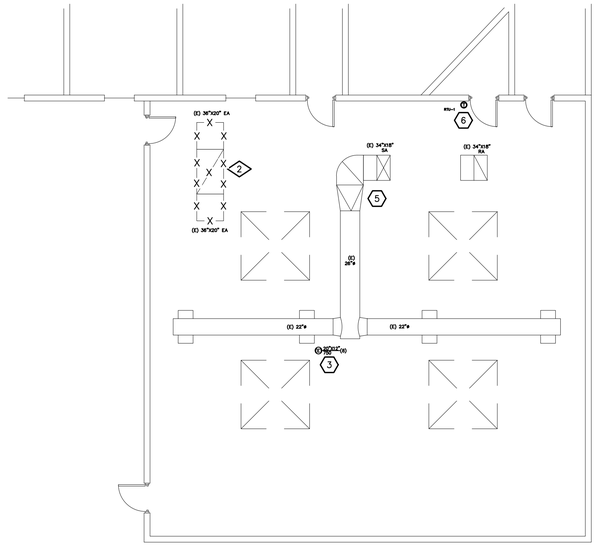
1. EXISTING ROOF TOP UNIT (RTU) TO BE DEMOLISHED. DISCONNECT BUILDING AUTOMATION CONTROLS, GAS LINE AND DUCTWORK CONNECTIONS, EXISTING CURB GASKETING AND RELATED EQUIPMENT ABOVE ROOF. DISPOSE OF UNIT. PROTECT ALL DUCT AND PIPE OPENINGS AND EXISTING CURB DURING REPLACEMENT. INSPECT CONDITION OF EXISTING CURB AND REPAIR/REPLACE AS NECESSARY TO RECEIVE NEW RTU.
2. EXISTING RELIEF HOOD AND ASSOCIATED DUCTWORK AND CONTROLS TO BE DEMOLISHED. DISCONNECT AND CAP ROOF OPENING ABOVE AND BELOW ROOF. PROVIDE INSULATED ROOF CAP TO PREVENT CONDENSATION, SLOPED FOR DRAINAGE, AND FLASHED FOR MOISTURE PROTECTION.

MECHANICAL WORK NOTES: (X)

1. EXAMINE EXISTING MECHANICAL UNIT CURB FOR STRUCTURAL INTEGRITY AND REUSE. REPAIR OR REPLACE DAMAGED CONDITIONS AS NEEDED. INSTALL NEW CURB ADAPTOR ON EXISTING CURB TO RECEIVE NEW RTU. SEAL ALL JOINTS WATER TIGHT. PROVIDE FIELD REINFORCEMENT OF THE EXISTING CURB AND/OR ROOF STRUCTURE AS REQUIRED.
2. INSTALL CURB ADAPTOR PER MANUFACTURER'S REQUIREMENTS. SEAL ALL JOINTS WATER TIGHT. ENSURE THE OUTSIDE AIR INTAKE IS LOCATED GREATER THAN 10 FEET HORIZONTALLY OF ANY EXHAUST DISCHARGE TERMINALS. INSTALL NEW RTU ON CURB ADAPTOR.
3. MAINTAIN MANUFACTURERS RECOMMENDED CLEARANCE REQUIREMENTS FOR MAINTENANCE, NOTING THAT TRANE COMPANY WILL ALLOW A REDUCE CLEARANCE MINIMUM OF 18" ON THE SKYLIGHT SIDE OF THE UNIT, SUBJECT TO NEC CLEARANCES BEING MET. SEE ADDITIONAL INSTALLATION REQUIREMENTS ON THESE DRAWINGS. THE NEW UNIT WITH CURB ADAPTOR WILL OVERHANG THE EXISTING ROOF CURB. REVIEW OVERHANG DIMENSIONS AND PROVIDE ANY STRUCTURE SUPPORT NEEDED.
4. LIMITED CLEARANCE FROM EXISTING SKYLIGHT. PROVIDE FALL PROTECTION SYSTEM INCLUDING PERSONAL FALL ARREST SYSTEM AND SKYLIGHT GUARD AS REQUIRED BY IBC/OSHA.
5. EXISTING DUCTWORK AND DIFFUSERS TO REMAIN. CLEAN EXISTING SYSTEMS TO REMOVE DEBRIS. REBALANCE AIR SYSTEM TO CFMS SHOWN ON THIS DRAWING. REPAIR OR REPLACE ANY DEFECTIVE PARTS TO FULLY OPERATIONAL.
6. COORDINATE LOCATION OF SPACE TEMPERATURE SENSOR WITH OWNER. PROVIDE TAMPER-PROOF, VANDAL RESISTANT ENCLOSURE.

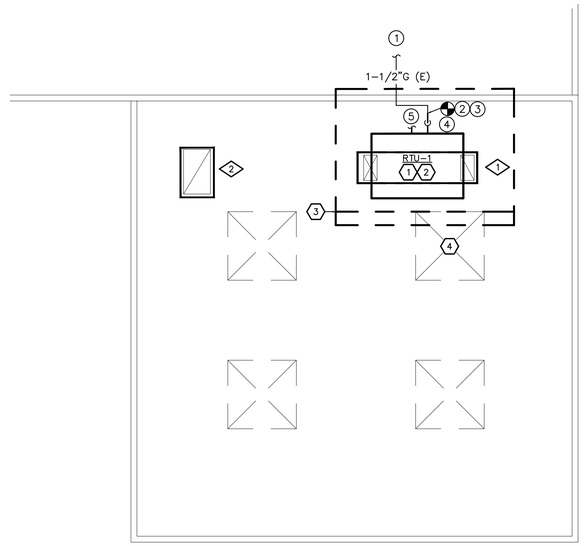
PLUMBING WORK NOTES: (X)

1. COORDINATE WITH GAS UTILITY TO CONFIRM THAT EXISTING GAS SERVICE CAN BE REUSED. CONTRACTOR TO FIELD VERIFY GAS LOAD REQUIREMENTS.
2. CONNECT NEW 1-1/2" GAS LINE FOR NEW RTU TO EXISTING GAS PIPING.
3. PRESSURE TEST, VISUALLY INSPECT AND PURGE ALL NEW GAS PIPING, AND COMPONENTS, IN ACCORDANCE WITH IFGC SECTION 406. CONTRACTOR SHALL FIELD VERIFY THE EXISTING PIPING SYSTEMS ARE SUFFICIENT AND CODE COMPLIANT FOR THE SUPPLY PRESSURE OF THE GAS DELIVERY SYSTEM AND REPORT ANY DISCREPANCIES TO THE ENGINEER. MAKE REQUIRED REPAIRS AND UPGRADES, TO EXISTING PIPING AND COMPONENTS, AS NECESSARY TO ENSURE COMPLIANCE. THIS NOTE APPLIES TO ALL GAS SYSTEM COMPONENTS AND PIPING RELATED TO THIS INSTALLATIONS, NEW AND EXISTING. REGARDLESS OF WHETHER THEY ARE SHOWN ON THIS PLAN OR ARE NOT SHOWN ON THIS PLAN AS NOT ALL COMPONENTS, PIPING AND APPLIANCES ARE SHOWN ON THIS PLAN. FIELD VERIFY ALL EXISTING COMPONENTS, GAS PIPING, AND APPLIANCES.
4. INSTALL AND CONNECT NEW 1-1/2" GAS LINE TO NEW ROOF TOP UNIT, SEE CONNECTION DETAIL FOR ADDITIONAL REQUIREMENTS. FIELD VERIFY EXACT REQUIREMENTS AND ROUTING.
5. CONDENSATE DRAIN 1" SLIP. SEE CONNECTION DETAIL FOR ADDITIONAL REQUIREMENTS.



2 / M1.1 MECHANICAL AND PLUMBING FLOOR PLAN
SCALE: 1/8"=1'-0"

NORTH



1 / M1.1 MECHANICAL AND PLUMBING ROOF PLAN
SCALE: 1/8"=1'-0"

NORTH

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COLORADO REGISTERED PETER C. DAVIDSON 33293 07/03/26 PROFESSIONAL ENGINEER

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GENERAL CONSTRUCTION NOTES:

1. THE ELECTRICAL CONTRACTOR SHALL VERIFY THAT ALL ELECTRICAL ITEMS TO REMAIN OR BE RELOCATED AND REUSED ARE IN WORKING ORDER PRIOR TO ANY DEMOLITION WORK. IF THE EXISTING MATERIAL IS FOUND TO BE INOPERABLE, CONTRACTOR SHALL INFORM THE OWNER. ONCE ANY DEMOLITION WORK HAS BEGUN, ANY INOPERABLE OR DAMAGED MATERIAL SHALL BE REPAIRED OR REPLACED AT THE CONTRACTOR'S EXPENSE.
2. VERIFICATION OF EXISTING CONDITIONS. "INASMUCH AS THE REMODELING AND/OR REHABILITATION OF THE EXISTING BUILDING REQUIRES THAT CERTAIN ASSUMPTIONS BE MADE REGARDING EXISTING CONDITIONS, AND BECAUSE SOME OF THESE ASSUMPTIONS MAY NOT BE VERIFIABLE WITHOUT DESTROYING OTHERWISE ADEQUATE OR SERVICEABLE PORTIONS OF THE BUILDING, THE GENERAL CONTRACTOR AGREES THAT, EXCEPT FOR NEGLIGENCE ON THAT PART OF THE DESIGN PROFESSIONAL THE CONTRACTOR WILL HOLD HARMLESS, INDEMNIFY AND DEFEND THE DESIGN PROFESSIONAL FROM AND AGAINST ANY AND ALL CLAIMS ARISING OUT OF THE PROFESSIONAL SERVICES PROVIDED."
3. ANY ELECTRICAL ITEMS SHOWN OR NOT SHOWN ON THE PLANS, OR WHERE CIRCUITS IN EXISTING WALLS ARE REMOVED BY DEMOLITION, SHALL UPON COMPLETION OF REMODEL WORK BE LEFT IN WORKING CONDITION.
4. ALL PHASES OF THE ELECTRICAL WORK SHALL BE COORDINATED WITH THE ARCHITECT. WORK SHALL BE DONE IN A FASHION TO CAUSE AS LITTLE INCONVENIENCE AS POSSIBLE TO THE OWNER.
5. ELECTRICAL DEVICES NOTED TO BE REMOVED SHALL BE REMOVED BACK TO A POINT WHERE EXISTING CONDUIT CAN BE ABANDONED IN CONCEALED SPACES. REMOVE ALL WIRING FROM ABANDONED CONDUIT. ALL BOXES TO BE REMOVED SHALL BE TAKEN OUT OF WALLS AND HAVE HOLES REFINISHED TO MATCH WALL FINISH.
6. ELECTRICAL CONTRACTOR SHALL NOT DEFACE ANY AREAS OF THE BUILDING WHERE REMODELING IS NOT BEING DONE.
7. ROUTING OF EXISTING CONCEALED CONDUIT NOT KNOWN. LOCATION DETERMINED BY ELECTRICAL CONTRACTOR. ELECTRICAL CONTRACTOR SHALL RECIRCUIT AS NOTED UTILIZING ANY EXISTING CONDUIT. HE SHALL REMOVE EXISTING WIRE AND REPULL NEW. ALL NEW CONDUIT ADDED SHALL BE CONCEALED WHEREVER POSSIBLE.
8. SURFACE RACEWAY: WHEREVER CONCEALED CONDUIT IN FINISHED AREAS IS NOT POSSIBLE, ELECTRICAL CONTRACTOR SHALL INSTALL SURFACE MOUNTED RACEWAYS EQUAL TO WIREMOLD. RUN SURFACE RACEWAYS IN CORNER OF WALL AND CEILING. ALL RACEWAYS THAT ARE EXPOSED SHALL BE APPROVED BY ARCHITECT PRIOR TO ROUGH-IN.
9. TERMINATING AND SPLICING: MAKE ALL JOINTS AND SPLICES IN BRANCH CIRCUIT WIRING WITH APPROVED SOLDERLESS TOOL APPLIED OR TWIST-ON CONNECTORS, IN THE VARIOUS BOXES, GUTTERS, AND SIMILAR LOCATIONS, BUT NOT IN RACEWAYS. LEAVE SUFFICIENT SLACK TO PERMIT TWO (2) OR MORE SPLICES OR JOINTS TO BE REMADE IN CASE OF FAULT.
10. NM (ROMEX CABLE) WILL NOT BE ALLOWED ON THIS PROJECT.
11. ELECTRICAL CONTRACTOR SHALL RECEIVE, FROM SYSTEM SUPPLIERS, ALL WIRING DIAGRAMS FOR ALL EQUIPMENT, PRIOR TO ANY ROUGH-IN, TO ASSURE PROPER ELECTRICAL CHARACTERISTICS ARE PROVIDED. ELECTRICAL CONTRACTOR SHALL PROVIDE ARCHITECT WRITTEN NOTIFICATION PRIOR TO ROUGH-IN, THAT ALL WIRING DIAGRAMS HAVE BEEN RECEIVED AND REVIEWED FOR CORRECTNESS. ANY INCORRECT WIRING OR DEVICES INSTALLED BY ELECTRICAL CONTRACTOR WITHOUT WIRING DIAGRAMS SHALL BE CORRECTED AT ELECTRICAL CONTRACTOR'S EXPENSE.
12. ELECTRICAL CONTRACTOR SHALL RECEIVE, FROM MECHANICAL CONTRACTOR, ALL WIRING DIAGRAMS AND SHOP DRAWINGS FOR ALL MECHANICAL EQUIPMENT, PRIOR TO ANY ROUGH-IN, TO ASSURE PROPER ELECTRICAL CHARACTERISTICS, VOLTAGE, PHASE, HORSEPOWER, AMPERE, KILOWATTS AND ETC. ARE PROVIDED. ELECTRICAL CONTRACTOR SHALL PROVIDE ARCHITECT WRITTEN NOTIFICATION PRIOR TO ANY ROUGH-IN, THAT ALL WIRING DIAGRAMS AND SHOP DRAWINGS HAVE BEEN RECEIVED AND REVIEWED FOR CORRECTNESS. ANY INCORRECT WIRING OR DEVICES INSTALLED BY ELECTRICAL CONTRACTOR WITHOUT WIRING DIAGRAMS SHALL BE CORRECTED AT ELECTRICAL CONTRACTOR'S EXPENSE.
13. ELECTRICAL CONTRACTOR SHALL VISIT THE SITE PRIOR TO BIDDING TO DETERMINE EXACT DEMOLITION WORK TO BE DONE AND SHALL INCLUDE ALL DEMOLITION COSTS IN THEIR BID.
14. ELECTRICAL CONTRACTOR SHALL VERIFY FINAL LOCATIONS OF ALL SINKS WITH PLUMBING CONTRACTOR PRIOR TO ROUGH-IN. ANY ELECTRICAL DEVICES LOCATED ABOVE COUNTER AND BEHIND FINAL SINK LOCATIONS SHALL BE SHIFTED A MINIMUM OF 6" TO EITHER SIDE OF SINK. ANY ELECTRICAL DEVICES LEFT BEHIND SINK AT TIME OF FINAL ELECTRICAL OBSERVATION SHALL BE RELOCATED AT ELECTRICAL CONTRACTOR'S EXPENSE.
15. RELOCATIONS: OWNER RESERVES THE RIGHT TO RELOCATE ANY ELECTRICAL DEVICE, UP TO A DISTANCE OF 12'-0", BEFORE INSTALLATION WITHOUT EXTRA CHARGE FROM ELECTRICAL CONTRACTOR.
16. ELECTRICAL DRAWINGS ARE DIAGRAMMATIC ONLY. EXACT LOCATION OF ALL SYSTEMS AND EQUIPMENT SHALL BE FIELD VERIFIED AND COORDINATED WITH OTHER TRADES PRIOR TO ANY INSTALLATION. WHERE EXACT LOCATIONS ARE NECESSARY, THEY ARE DIMENSIONED ON THESE DRAWINGS. WHERE THERE IS A QUESTION OF ADEQUATE CLEARANCE OR COORDINATION BETWEEN TRADES, THIS CONTRACTOR SHALL PREPARE SHOP DRAWINGS FOR ENGINEER'S REVIEW. ON ALL SPECIAL SYSTEMS REQUIRING DRAWINGS BY LICENSED INSTALLATION CONTRACTORS, SUCH AS FIRE PROTECTION, SUCH DRAWINGS SHALL BE SUBMITTED WITHIN 30 DAYS AFTER AWARD OF CONTRACT.
17. CALL UTILITY COMPANIES (POWER, GAS, WATER, SEWER, TELEPHONE, CABLE TV, ETC.) IN ADVANCE BEFORE TRENCHING FOR THE MARKING OF THEIR UNDERGROUND UTILITIES. ALSO CONTRACTOR SHALL LOCATE ALL ON-SITE UTILITIES SUCH AS SECONDARY SERVICE FEEDERS, UNDERGROUND ELECTRICAL BRANCH CIRCUITS, SPRINKLER LINES, ETC. PRIOR TO TRENCHING. ANY CUT OR DAMAGED UNDERGROUND UTILITIES SHALL BE REPAIRED OR REPLACED AT CONTRACTOR'S EXPENSE.
18. EMT CONDUIT FITTINGS: DRY LOCATIONS ALL EMT COUPLERS AND CONNECTORS SHALL BE STEEL SET SCREW TYPE. DIE CAST FITTINGS SHALL NOT BE USED ON THIS PROJECT. DAMP/WET LOCATIONS, USE STEEL COMPRESSION GLAND TYPE COUPLER AND CONNECTORS.
19. ACCESS PANELS REQUIRED BY THE ELECTRICAL CONTRACTOR SHALL BE PROVIDED BY THE ELECTRICAL BID CONTRACTOR, THEN TURNED OVER TO THE APPROPRIATE TRADE FOR INSTALLATION. SEE ARCHITECTURAL SPECIFICATION SECTION DIVISION 8.
20. CONTRACTOR SHALL NOT FASTEN, ATTACH OR HANG ANY MATERIAL FROM THE ROOF DECK. ALL CONDUITS, JUNCTION BOXES, FIXTURES, DEVICES AND EQUIPMENT SHALL BE HUNG FROM THE STRUCTURAL STEEL FRAME AND SHALL BE PLACED WITH A MINIMUM OF 1" CLEARANCE BELOW THE ROOF DECK. WIRING AND CONDUITS SHALL NOT BE PLACED WITHIN THE RIBS OF THE ROOF DECK. CONTRACTOR SHALL NOT LOOSEN, REMOVE OR CUT ANY ROOFING SYSTEM FASTENERS PROTRUDING THROUGH THE ROOF DECK.

ELECTRICAL NOTES:

1. PROVIDE ALL EQUIPMENT AND SERVICES NECESSARY FOR COMPLETE SYSTEM.
2. EQUIPMENT AND MATERIALS SHALL BE UL-LISTED FOR THE USE INTENDED.
3. SUPPORT CONDUIT AND EQUIPMENT FROM THE STRUCTURE TO PREVENT SAGGING, POCKETING, AND VIBRATION.
4. FIRE-CAULK ALL PIPING PASSING THROUGH FIRE-RATED ASSEMBLIES.
5. PROVIDE COMPLETE SYSTEMS OF CONDUCTORS AND RACEWAYS USING CONDUIT AND/OR CABLE ASSEMBLIES APPROPRIATE TO THE FUNCTION AND LOCATION AND APPROVED PER NEC REQUIREMENTS. CONDUCTORS MUST BE DERATED PER CODE.
6. PROVIDE A COMPLETE GROUNDING SYSTEM PER NEC, SECTION 250.
7. RACEWAYS MAY BE THE FOLLOWING (SUBJECT TO NEC REQUIREMENTS): ELECTRICAL METALLIC TUBING, GALVANIZED.
8. PROVIDE GALVANIZED STEEL JUNCTION AND OUTLET BOXES, EXCEPT AS INDICATED OTHERWISE. PROVIDE WEATHER-PROOF TYPE CAST BOXES WITH GASKET AND CAST COVERPLATE FOR WET LOCATIONS. THROUGH THE WALL BOXES ARE NOT PERMITTED.
9. POWER CONDUCTORS SHALL BE COPPER, (STRANDED FOR NO. 8 AWG AND LARGER), WITH THE FOLLOWING INSULATION STANDARDS: THHN, THWN, XHHW (90C). THE CONDUCTORS SHALL BE APPLIED USING THE 75C RATING.
10. PROVIDE SPECIFICATION GRADE, IVORY COLORED DEVICES THROUGHOUT, UNLESS NOTED OTHERWISE ON THE PLANS. DEVICES SHALL BE HUBBELL. PLATES SHALL BE IVORY PLASTIC, WITH MATCHING SCREWS, UNLESS NOTED OTHERWISE. PROVIDE BLANK COVERPLATES FOR UNUSED OUTLETS. COVERPLATES FOR MULTI-GANG BOXES SHALL BE SIZED FOR THE BOX T COVERS.
11. BRANCH CIRCUITS SHALL USE MINIMUM NO. 12 AWG WIRING. CONTROL WIRING MINIMUM NO. 14 AWG. FOR DISTANCES EXCEEDING 75' FROM THE PANEL, PROVIDE MIN. 10 AWG.
12. ALL CONDUIT AND DEVICES SHALL BE SURFACE MOUNTED. REFER TO PLAN AND PLAN NOTES.
13. RUN ALL CONDUIT PARALLEL AND PERPENDICULAR TO BUILDING LINES.
14. USE FLEXIBLE METALLIC CONDUIT FOR CONNECTIONS TO MOTORS, FIXTURES, OR OTHER EQUIPMENT WHERE VIBRATION IS PRESENT. PROVIDE SEALTITE FMC IN WET AREAS.
15. PROVIDE GROUND WIRE IN NON-METALLIC CONDUIT AND FLEXIBLE CONDUIT. GROUND WIRES SHALL BE INCREASED IN SIZE WHERE CIRCUIT WIRING IS INCREASED FOR VOLTAGE DROP.
16. PROVIDE A TYPED PANEL SCHEDULE FOR EACH PANEL, INSERTED IN A PLASTIC SLEEVE.

ELECTRICAL LEGEND

SYMBOL	DESCRIPTION
	MAIN DISTRIBUTION BOARD
	PANELBOARD
M	ELECTRIC SERVICE METER
CT	CURRENT TRANSFORMER
	NEW WORK UNDER THIS CONTRACT
	WORK DEMOLISHED UNDER THIS CONTRACT
	WORK EXISTING TO REMAIN UNDER THIS CONTRACT
	CONCEALED CIRCUIT
	UNDERGROUND/FLOOR CIRCUIT
	EXPOSED CIRCUIT
WM	WIREMOLD (SURFACE RACEWAY)
PM	PLUGMOLD (SURFACE RACEWAY)
P – 1,3,5	HOMERUN TO PANELBOARD (ONE ARROW / CIRCUIT, PANEL AND CIRCUIT #'S SHOWN)
	CONDUIT TURNS UP
	CONDUIT TURNS DOWN
J	JUNCTION BOX
M	MOTOR (REFER TO MOTOR SCHEDULE)
	SAFETY DISCONNECT SWITCH (REFER TO MOTOR SCHEDULE)
	RECEPTACLE - DUPLEX
	RECEPTACLE - DUPLEX WITH ISOLATED GROUND
	RECEPTACLE - DUPLEX WITH GFCI PROTECTION
	RECEPTACLE - DUPLEX SPLIT WIRE
	RECEPTACLE - QUADRAPLEX (DOUBLE DUPLEX)
	RECEPTACLE - SPECIAL PURPOSE - SINGLE PHASE
	RECEPTACLE - SPECIAL PURPOSE - THREE PHASE
	RECEPTACLE - SINGLE OUTLET -
	RECEPTACLE - RANGE
	RECEPTACLE - DUPLEX - FLOOR MOUNTED

SYMBOL	DESCRIPTION
	SURFACE MOUNT 1x4 LUMINAIRE
	SURFACE MOUNT 2x4 LUMINAIRE
	SURFACE MOUNT 2x4 LUMINAIRE - UN-SWITCHED (NIGHT-LIGHT)
	SURFACE MOUNT ROUND LUMINAIRE
	SURFACE MOUNT STRIP LUMINAIRE
	WALL MOUNT STRIP LUMINAIRE
	RECESSED 2x4 LUMINAIRE
	RECESSED 2x4 LUMINAIRE - UN-SWITCHED (NIGHT-LIGHT)
	RECESSED 2X4 LUMINAIRE WITH EMERGENCY BATTERY BACK-UP
	RECESSED 2x2 LUMINAIRE
	RECESSED 2x2 LUMINAIRE WITH EMERGENCY BATTERY BACK-UP
	RECESSED ROUND LUMINAIRE
	RECESSED ROUND LUMINAIRE WITH EMERGENCY BATTERY BACK-UP
	PENDANT LUMINAIRE
	WALL MOUNTED LUMINAIRE
	POLE MOUNTED LUMINAIRE
	WALL MOUNTED EMERGENCY BATTERY BACK-UP
	CEILING MOUNTED EMERGENCY BATTERY BACK-UP
	EXIT SIGN WITH DIRECTIONAL ARROWS HATCH INDICATES ILLUMINATED FACES, ARROWS INDICATE DIRECTIONS
a $ F	SWITCH (TOGGLE UNLESS NOTED): x - SWITCH GROUP "x" M - MANUAL WITH THERMAL OVERLOAD 2 - 2 POLE p - PILOT LIGHT 3 - 3-WAY k - KEYED 4 - 4-WAY t - TIMER D - DIMMER F - FUSTAT VS - VACANCY SENSOR V - VARIABLE SPEED ROMAN NUMERAL (I,II,III, etc) INDICATES QUANTITY OF GANGED DEVICES UNDER COMMON FACEPLATE
OS	OS - OCCUPANCY SENSOR WITH MANUAL OVERRIDE

SYMBOL	DESCRIPTION
	DATA OUTLET - 3/4" CONDUIT TO ABOVE CEILING.
	TV - SATELLITE TV OUTLET. 1.5" CONDUIT TO ABOVE CEILING.
	HDMI - HDMI OUTLET. 1.5" CONDUIT TO ABOVE CEILING.
XX	MISCELLANEOUS SYSTEMS DEVICE DB - DOORBELL CH - CHIME CK - CARD READER / KEYPAD CR - CARD READER DA - DURESS ALARM DC - DOOR CONTACT DE - DELAYED EGRESS EL - ELECTRIC LOCK ES - ELECTRIC STRIKE IC - INTERCOM CALL STATION KP - KEYPAD MD - MOTION DETECTOR ML - MAGNETIC LOCK RX - REQUEST TO EXIT

GENERAL ELECTRICAL ABBREVIATIONS

AC - ABOVE COUNTER
AFF - ABOVE FINISHED FLOOR
AFG - ABOVE FINISHED GRADE
BFC - BELOW FINISHED CEILING
BAS - BUILDING AUTOMATION SYSTEM
BLDG - BUILDING
BMS - BUILDING MANAGEMENT SYSTEM
BOF - BOTTOM OF FIXTURE
COF - COFFEE MAKER
CKT - CIRCUIT
CLG - CEILING
CO - CONDUIT ONLY
CWA - COORDINATE WITH ARCHITECT
CWM - COORDINATE WITH EQ SUPPLIER
DW - DISHWASHER
EF - EXHAUST FAN
EHD - ELECTRIC HAND DRYER
EM - EMERGENCY
EMS - ENERGY MANAGEMENT SYSTEM
EP - EXPLOSION PROOF
EWC - ELECTRIC WATER COOLER
EX - EXISTING
FRZ - FREEZER
FA - FIRE ALARM
FCP - FIRE ALARM CONTROL PANEL
GD - GARBAGE DISPOSAL
GND - GROUND
GFI - GROUND FAULT CIRCUIT INTERRUPTER
GFCI - GROUND FAULT CIRCUIT INTERRUPTER
HZL - HAZARDOUS LOCATION
IHW - INSTANT HOT WATER DISPENSER
JB - JUNCTION BOX
LGT(S) - LIGHT(S)
LV - LOW VOLTAGE
MAX - MAXIMUM
MGMT - MANAGEMENT
MIN - MINIMUM
MOP - MOTOR OPERATED PARITITION
MW - MICROWAVE
NC - NORMALLY CLOSED
NL - NIGHT LIGHT
NO - NORMALLY OPEN.
OVN - OVEN
PRJ - CEILING MOUNTED PROJECTOR
PS - ELECTRIC PROJECTOR SCREEN
PB - PULLBOX
RCPT(S) - RECEPTACLE(S)
RNG - RANGE
ST - SHUNT TRIP
SPD - SURGE PROTECTIVE DEVICE
REF - REFRIGERATOR
TV - TELEVISION
UNO - UNLESS NOTED OTHERWISE
UVR - UNDER-VOLTAGE RELEASE
UH - UNIT HEATER
VFD - VARIABLE FREQUENCY DRIVE
XFMR - TRANSFORMER
WP - WEATHERPROOF

NOTE: NOT ALL SYMBOLS OR ABBREVIATIONS ON THIS LEGEND ARE NECESSARILY USED ON THIS PROJECT.

INDEX OF ELECTRICAL DRAWINGS

SHEET	TITLE
E-01	SYMBOLS AND NOTES
E-10	ELECTRICAL PLAN

CAIRN DESIGN ENGINEERS ARCHITECTS
leading high performance building solutions

leading high performance building solutions

MWBE - DBE - SBE

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COLORADO REGISTERED KENNETH P. CAIRN 48202 PROFESSIONAL ENGINEER

PROJECT:

WELD COUNTY
JAIL NORTH
JAIL 1 EAST
MAKEUP AIR UNIT

SHEET TITLE:

SYMBOLS AND
SPECIFICATIONS

ISSUE DATE:

BID/PERMIT
07/03/2026

REVISIONS:

PROJECT NO.:
2607

SHEET NO.

E-01

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EXISTING H2E

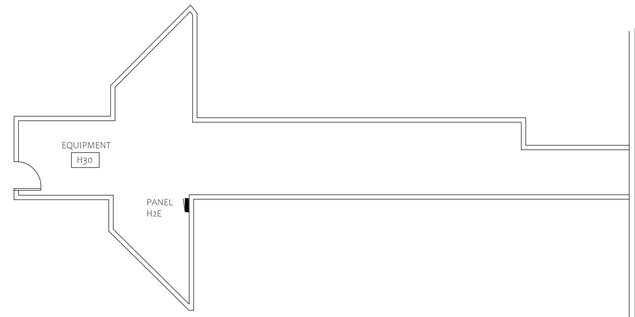
Location : ELEC
Supply From : MDC
Mounting : SURFACE
Voltage Line-Line : 277
Voltage Line-Ground : 480
A.I.C Rating (A) : 18000
Main Type : MLO

NEMA : 1
Phase : 3
Wires : 4
Main Rating (A) : 400A
MCB Rating (A) : N/A

Notes :
1. EXISTING BREAKER TO BE REPLACED.
2. PANEL IS FED FROM A 400A FUSED SWITCH.
3. PANEL IS SIZED FOR 400A.
4.
5.
6.

Notes	LOAD DESCRIPTION	SIZE (A)	POLES	CKT. No	VOLT-AMPS PER PHASE A	A	B	B	C	C	CKT. No	SIZE (A)	POLES	LOAD DESCRIPTION	Notes
		60		1	8484	3500					2	20	1	LIGHTS	
	RTU-23	60	3	3			8484	3500			4	20	1	LIGHTS	
		60		5					8484	3500	6	20	1	LIGHTS	
		20		7	3167	2575					8	20	1	LIGHTS	
	RTU-24	20	3	9			3167	2915			10	20	1	LIGHTS	
		20		11					3167	3230	12	20	1	LIGHTS	
		50		13	8225	2730					14	20	1	LIGHTS	
1	RTU-25 (NEW RTU-1)	50	3	15			8225	2575			16	20	1	LIGHTS	
		50		17					8225	3130	18	20	1	LIGHTS	
		15		19	400	2425					20	20	1	LIGHTS	
	TEF-24	15	3	21			400	1165			22	20	1	LIGHTS	
		15		23					400	2355	24	20	1	LIGHTS	
	UNIT HEATER	30	1	25	2000	3560					26	20	1	LIGHTS	
	UNIT HEATER	30	1	27			2000	2370			28	20	1	LIGHTS	
	UNIT HEATER	30	1	29					2000	2700	30	20	1	LIGHTS	
		20		31	800	2700					32	20	1	LIGHTS	
	TEF-25	20	3	33			800	0			34	20	1	SPARE	
		20		35					800	0	36	20	1	SPARE	
		60		37	12000	16400					38	90			
	UNIT HEATER	60	3	39			12000	16400			40	90	3	TRANSFORMER	
		60		41					12000	16400	42	90			
	Connected load per phase				68,966		64,001		66,391						

	Total Load		Demand		Calculated Load
Lighting load	44,930	x	1.25		56,163 VA
Receptacle load (10 kVA or less)	10,000	x	1.00		10,000 VA
Receptacle load over 10 kVA	60,405	x	0.50		30,203 VA
Largest motor	25,452	x	1.25		31,815 VA
Other motors	37,776	x	1.00		37,776 VA
Electric heat	42,000	x	1.25		52,500 VA
Other	0	x	1.00		0 VA
Fut1	0	x	0.37		0 VA
Fut2	0	x	0.37		0 VA
Fut3	0	x	0.37		0 VA
Total calculated load					270,956 VA
Minimum Panel Ampacity					326 Amps

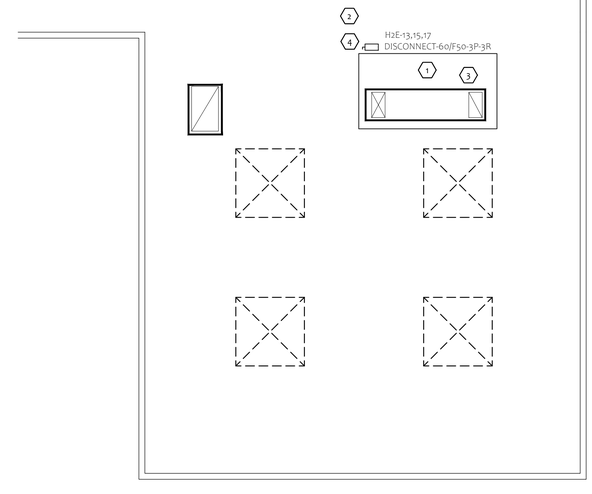


GENERAL NOTES:

1. THE CONTRACTOR SHALL VERIFY ALL EQUIPMENT BEING PROPOSED FITS IN THE LOCATIONS SHOWN PRIOR TO ORDERING EQUIPMENT. IF THE PROPOSED EQUIPMENT DOES NOT FIT, THE CONTRACTOR SHALL RECOMMEND AN ALTERNATE LOCATION FOR APPROVAL.

WORK NOTES:

1. NEW MAKEUP AIR UNIT. 480V, 3Ø, 43MCA, 50MOCP. UNIT IS PROVIDED WITH SUPPLY AND RETURN SMOKE DETECTORS. ELECTRICAL CONTRACTOR SHALL CONTRACT WITH A FIRE ALARM CONTRACTOR TO CONNECT BOTH DETECTORS TO THE EXISTING FIRE ALARM SYSTEM.
2. PROVIDE NEW DISCONNECT AND FEEDER TO PANEL H2E. PROVIDE 3#8, 1#10G, $\frac{3}{4}$" C. EXISTING CONDUIT MAY BE USED IF APPLICABLE; HOWEVER, CONTRACTOR SHALL INCLUDE A NEW CONDUIT RUN IN THEIR SCOPE.
3. REUSE CONDUIT PENETRATIONS WHERE APPLICABLE, CAP ALL UNUSED CONDUITS WITH WEATHERPROOF FITTINGS.
4. REPLACE EXISTING 25A/3P BREAKER FEEDING RTU. PROVIDE AND INSTALL A 50A/3P, GE TED, 18K BREAKER IN SPACES NOTED. PANEL H2E IS APPROXIMATELY 100' FROM THE RTU ON THE 2ND FLOOR OF JAIL 1 EAST.



01 ELECTRICAL PLAN
SCALE: 1/8 = 1'-0"

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COLORADO REGISTERED PROFESSIONAL ENGINEER 48202

PROJECT:
WELD COUNTY
JAIL NORTH
JAIL 1 EAST
MAKEUP AIR UNIT

SHEET TITLE:
ELECTRICAL
NEW WORK PLAN

ISSUE DATE:
BID/PERMIT
07/03/2026

REVISIONS:

PROJECT NO.:
2607

SHEET NO.
E-10

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